

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Pharm. Examination

University Theory Examination November 2016

PH319 Pharmaceutical Microbiology and Biotechnology

Date: 23.11.2016, Wednesday Time: 10:00 a.m. to 1:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. 20

1. The Process of alternate heating and cooling to kill the microbes present in milk is called as :
[A] Boiling
[B] Tyndallization
[C] Pasteurization
[D] Steam sterilization
2. The major constituents of agar are
[A] Fats
[B] Aminoacids
[C] Polypeptides
[D] Polysaccharides
3. _____ involves enclosing the enzymes within semi-permeable polymer membranes in the shape of spheres.
[A] Covalent Binding
[B] Encapsulation
[C] Adsorption
[D] None of the above
4. The strain of fungi used for the large scale production of penicillin is
[A] Penicillium chrysogenum
[B] Penicillium notatum
[C] Saccharomyces species
[D] Streptomyces Aureus
5. In sterility Testing, for the identification of aerobic bacteria and lower fungi, _____ medium is used.
[A] Fluid-thioglycolate

- [B] Soyabean casein digest
[C] Peptone broth
[D] Cooked meat
6. Following is not a method of evaluation of disinfection.
[A] In-use Test
[B] Suspension Test
[C] Fumigation
[D] Carrier Test
7. The generation time is
[A] The time required for the cell to divide
[B] The total division of the cell during its life time
[C] The total number of cells formed
[D] None of the above
8. _____ is an example of toxoid vaccine.
[A] Polio vaccine
[B] Measles vaccine
[C] Hepatitis A vaccine
[D] Tetanus vaccine
9. Natural polymer used in occlusion process of enzyme immobilization is
[A] Starch
[B] Polyvinylpyrrolidone
[C] Polyvinyl alcohol
[D] Polyacrylamide
10. Fusion from mixed populations of protoplasts is greatly enhanced by the use of _____.
[A] Propyl Alcohol
[B] Polyvinyl Pyrrolidone
[C] Propylene Glycol
[D] Polyethylene glycol
11. Capsule of bacteria do not accept most of chemical dyes because of their chemical composition so it appears surrounding each bacteria as
[A] Endospore
[B] Halos
[C] Spikes
[D] None of the above
12. Escherichia Coli is an example of
[A] Non-halophile
[B] Extreme halophile
[C] Halophile
[D] Halotolerant
13. Magnification range of light microscope is
[A] 1000x – 5000x
[B] 1000x – 2000x
[C] 500x – 1000x
[D] None of these

14. The viruses that live as parasites on bacteria are
[A] Bacteriophages
[B] Algae
[C] Fungi
[D] None of these
15. Ethanol is produced as result of fermentation process by _____ microorganism.
[A] *Bacillus subtilis*
[B] *Saccharomyces cerevisiae*
[C] *Lactobacillus bulgaricus*
[D] *Clostridium acetobutylicum*
16. During conjugation the genetic material is getting transferred through
[A] Pili
[B] Capsule
[C] Medium
[D] Cell wall
17. The enzyme which reduces oxygen to water for the cultivation of anaerobic bacteria is
[A] Lipase
[B] Amylase
[C] Peptidase
[D] Oxyrase
18. The Small segment of DNA that can move from one region of a DNA molecule to another (Same chromosome or on another chromosome) is called as
[A] Protoplast fusion
[B] Gene coding
[C] Recombination
[D] Transposons
19. Autoclave sterilize the product by
[A] Dry Heat Sterilization method
[B] Moist Heat Sterilization method
[C] Gas Sterilization method
[D] All of the above
20. Which of the following is the characteristic of virus structure?
[A] ATP generating metabolism
[B] Plasma Membrane
[C] Sensitivity to Interferon
[D] Ribosome

SECTION – II

Q 2 Attempt any **TWO** of the following

- A** Enumerate the Microbiological assay techniques of antibiotics and describe any one method in detail. **05**

- B** Define: Microbial Biotransformation. How household vinegar can be produced by the submerged fermentor? **05**
- C** Describe in detail the various phases of 'Microbial Growth Cycle'. **05**
- Q 3** Attempt any **TWO** of the following
- A** Classify the various methods for microbial biotransformation and distinguish the salient features of 'Growing Culture' process. **05**
- B** Compare and Contrast: Selective media and differential media with suitable examples. **05**
- C** Apply the principle of fermentation for the manufacturing of penicillin by means of flowchart. **05**

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Discuss briefly the methods used for measuring the effectiveness of disinfectants. **05**
- B** Write a brief note on Gene cloning process and its application. **05**
- C** Describe the various features and characteristics of 'Spirochetes'. **05**
- D** Write a brief note on different medias used for sterility testing. **05**
- E** Write a note on 'Conjugation' as a method of recombinant DNA technology. **05**
- F** Write a note on different carrier binding techniques used for enzyme immobilization. **05**
- Q 5** Attempt any **FOUR** of the following
- A** Enlist the various staining techniques for micro-organisms and explain in brief 'Zeil Neilsson staining technique'. **05**
- B** Define sterilization and discuss in detail dry heat sterilization process. **05**
- C** Write a brief note on pour plate and spread plate as techniques for determining microbial content in samples. **05**
- D** What do you mean by sterility testing? Describe the 'membrane filtration' as a method for sterility testing. **05**
- E** Give the detailed classification of disinfectants and discuss in brief quaternary ammonium compounds as disinfectants. **05**
- F** Discuss the process of protoplast fusion and its application. **05**